KKR & KSR INSTITUTE OF TECHNOLOGY AND SCIENCES

(Autonomous)
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
Semester – IV-I

Course Code	Course Name	L T P C
	CELLULAR & MOBILE COMMUNICATIONS	3 0 0 3

COURSE OUTCOMES:

CO1 : To Identify the limitations of conventional mobile Telephone systems and to Understand the concepts of cellular systems
CO2 : To understand the different types of interferences influencing cellular and mobile communications
CO3 : To understand the frequency management, channel assignment and types of handoffs
CO4 : To Understand the concept of handoff in cellular systems
CO5 : To understand the Multiple access schemes in mobile and to develop the ability to search, select, organize and present information on new technologies in mobile and cellular communications

UNIT-I: CELLULAR SYSTEMS

Limitations of Conventional System, Basic Cellular Mobile System, First, second, third and fourth and Generation cellular wireless systems. Operation of Cellular System. Uniqueness of Mobile Radio Environment –Fading, coherence bandwidth, Doppler Spread.
Fundamentals of cellular Radio System Design: concept of frequency reuse channels, Co-channel Interference, Co-channel Interference Reduction Factor, desired C/I from a normal case in a Omni directional Antenna system. Trunking and grade of service

UNIT-II: CO-CHANNEL & NON-CO-CHANNEL INTERFERENCE

Measurement of Real Time Co-Channel Interference, design of Antenna system, Antenna parameters and their effects, diversity techniques: Space Diversity, Polarization diversity, frequency diversity and time diversity.
Non-co channel interference: adjacent channel interference, Near End far end interference, effect on coverage and interference by power decrease, antenna height decrease

UNIT-III: FREQUENCY MANAGEMENT AND CHANNEL ASSIGNMENT & CELL COVERAGE FOR SIGNAL AND TRAFFIC

FREQUENCY MANAGEMENT AND CHANNEL ASSIGNMENT: Numbering and grouping, setup access and paging channels, channel assignments to cell sites and mobile units, channel sharing and borrowing, sectorization, overlaid cells, non-fixed channel assignment.
CELL COVERAGE FOR SIGNAL AND TRAFFIC: Signal reflections in flat and hilly terrain, effect of human made structures, phase difference between direct and reflected paths, straight line path loss slope, general formula for mobile propagation over water and flat open area, near and long distance propagation, antenna height gain, form of a point to point model.

UNIT-IV: CELLSITE AND MOBILE ANTENNAS & HANDOFFS

CELLSITE AND MOBILE ANTENNAS: Sum and difference patterns and their synthesis, omni directional antennas, directional antennas for interference reduction, space diversity antennas, umbrella pattern antennas, minimum separation of cell site antennas, high gain antennas.
HANDOFFS: Handoff Initiation, types of handoffs, delaying handoff, advantages of Handoff, power difference handoff, forced handoff, mobile assisted and soft and Hard handoffs. Intersystem and Intrasystem handoffs, dropped call rates and their evaluation.

UNIT-V: DIGITAL CELLULARNETWORKS AND MULTIPLE ACCESS SCHEMES

Digital cellular networks and multiple access schemes gsm architecture, GSM channels, TDMA, FDMA, and CDMA.Introduction to MIMO systems Principles of CDMA cellular systems, Principles of OFDMbased broadband wireless systems, 4G LTE basics – OFDM, and OFDMA, Generalized framework for Filtered OFDM and FBMC, Introduction of 5G.

TEXT BOOKS:

1. Mobile Cellular Telecommunications – W.C.Y. Lee, Tata McGraw Hill, 2rd Edn., 2006. Wireless Communications - Theodore. S. Rapport, Pearson education, 2nd Edn., 2002.
2. T. S. Rappaport, “Wireless Communications – Principles and Practice” (2nd edition) Pearson, 2010, ISBN 9788131731864

REFERENCE BOOKS:

1. Principles of Mobile Communications – Gordon L. Stuber, Springer International 2nd Edition, 2001.
2. Modern Wireless Communication –Simon Haykin Michael Moher, Persons Eduction,2005. 3. Wireless Communication theory and Techniques, Asrar U.H .Sheikh, Springer,2004.
3. A. Goldsmith, “Wireless Communications,” Cambridge Univ Press, 2005
4. D. Tse and P. Viswanath, “Fundamentals of Wireless Communications,” Cambridge Univ Press, 2005

KKR & KSR INSTITUTE OF TECHNOLOGY AND SCIENCES

(Autonomous)
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
Semester - VII

Course Code	Course Name	L T P C
(PE-V)	DIGITAL IMAGE PROCESSING	3 0 0 3

COURSE OUTCOMES:

CO1 : Describe the Image Processing system, scope of digital image processing and compare various image transforms.
CO2 : Apply filtering operations on images both in spatial and frequency domain.
CO3 : Describe Image degradation model and Explain the restoration techniques on images.
CO4 : Analyze the digital Images using wavelets and multi resolution processing and use various coding techniques for various image compression methods.
CO5 : Explain various segmentation techniques on digital images; Explain various Colour models and Colour Image Processing.

UNIT-I :

Introduction: Introduction to Image Processing, Examples of fields that use Digital Image Processing, Fundamental steps in digital image processing, components of an image processing system, image sampling and quantization, Some basic relationships between pixels.
Image Transforms: Need for image transforms, Image transforms, Fourier Transform, 2D Discrete Fourier Transform and its properties, Walsh Transform, Hadamard transform, Haar Transform, Discrete Cosine transform, KL Transform, Singular Value Decomposition.

UNIT-II :

Intensity Transformations and Spatial Filtering: Background, Some basic intensity transformation functions, histogram processing, fundamentals of spatial filtering, smoothing spatial filters, and sharpening spatial filters.
Filtering in the Frequency Domain: The Basics of filtering in the frequency domain, image smoothing using frequency domain filters, Image Sharpening using frequency domain filters, Selective filtering.

UNIT-III :

Image Restoration and Reconstruction: A model of the image degradation / Restoration process, Noise models, restoration in the presence of noise only-Spatial Filtering, Periodic Noise Reduction by frequency domain filtering, Linear, Position –Invariant Degradations, Estimating the degradation function, Inverse filtering, Minimum mean square error (Wiener) filtering.

UNIT-IV :

Wavelets and Multi resolution Processing: Image pyramids, sub band coding, Multi resolution expansions, wavelet transforms in one dimensions& two dimensions, Wavelet packets.
Image compression: Fundamentals, Basic compression methods: Huffman coding, Golomb coding, Arithmetic coding, LZW coding, Run-Length coding, Block Transform coding, Predictive coding.

UNIT-V :

Image segmentation: Fundamentals, point, line, edge detection, thresholding, and Region – based segmentation.
Color image processing: Color fundamentals, color models, pseudo color image processing, basics of full color image processing, color transformations, smoothing and sharpening. Image segmentation based on color, noise in color images, color image compression.

TEXT BOOKS:

1. R. C. Gonzalez and R. E. Woods, “Digital Image Processing, 3rd edition, Pearson,2008.

REFERENCE BOOKS:

1. Anil K.Jain, “Fundamentals of Digital Image Processing”, Prentice Hall of India, 9th Edition, Indian Reprint, 2002.
2. Jayaraman, S. Esakkirajan, and T. Veerakumar, “Digital Image Processing”, Tata McGraw-Hill Education, 2009.

KKR & KSR INSTITUTE OF TECHNOLOGY AND SCIENCES

(Autonomous)
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
Semester – IV-I

Course Code	Course Name	L T P C
(PE-IV)	DSP PROCESSORS AND ARCHITECTURES	3 0 0 3

COURSE OUTCOMES:

CO1 : Understand the basics of Digital Signal Processing and transforms.
CO2 : Able to distinguish between the architectural features of general purpose processors and DSP processors.
CO3 : Understand the architectures of TMS320C54xx devices and ADSP 2100 DSP devices.
CO4 : Able to write simple assembly language programs using instruction set of TMS320C54xx.
CO5 : Can interface various devices to DSP Processors.

UNIT-I:

Introduction to Digital Signal Processing: Introduction, A Digital signal-processing system, The sampling process, Discrete time sequences, Discrete Fourier Transform (DFT) and Fast Fourier Transform (FFT), Linear time-invariant systems, Digital filters, Decimation and interpolation.
Computational Accuracy in DSP Implementations: Number formats for signals and coefficients in DSP systems, Dynamic Range and Precision, Sources of error in DSP implementations, A/D Conversion errors, DSP Computational errors, D/A Conversion Errors, Compensating filter.

UNIT-II:

Architectures for Programmable DSP Devices: Basic Architectural features, DSP Computational Building Blocks, Bus Architecture and Memory, Data Addressing Capabilities, Address Generation Unit, Programmability and Program Execution, Speed Issues, Features for External interfacing.

UNIT-III:

Programmable Digital Signal Processors: Commercial digital signal processing devices, Data Addressing modes of TMS320C54XX DSPs, data Addressing modes of TMS320C54XX Processors, Memory space of TMS320C54XX processors, program control, TMS320C54XX instructions and programming, On-Chip Peripherals, Interrupts of TMS320C54XX processors, pipeline Operation of TMS320C54XX Processors.

UNIT-IV:

Analog Devices Family of DSP Devices: Analog Devices Family of DSP Devices ALU and MAC block diagram, Shifter Instruction, Base Architecture of ADSP 2100, ADSP2181 high performance processor. Introduction to Blackfin Processor- The Blackfin Processor, Introduction to Micro signal Architecture, Overview of Hardware Processing Units and Register files, Address Arithmetic Unit, Control Unit, Bus Architecture and Memory, Basic Peripherals.

UNIT-V:

Interfacing Memory and I/O Peripherals to Programmable DSP Devices: Memory space organization, External bus interfacing signals, Memory interface, Parallel I/O interface, Programmed I/O, Interrupts and I/O, Direct memory access (DMA).

TEXT BOOKS:

1. Digital Signal Processing – Avtar Singh and S. Srinivasan, Thomson Publications, 2004.
2. A Practical Approach To Digital Signal Processing - K Padmanabhan, R. Vijayarajeswaran, Ananthi. S, New Age International, 2006/2009.
3. Embedded Signal Processing with the Micro Signal Architecture Publisher: Woon-SengGan, Sen M. Kuo, Wiley-IEEE Press, 2007.

REFERENCE BOOKS:

1. Digital Signal Processors, Architecture, Programming and Applications–B. Venkataramani and M. Bhaskar, 2002, TMH.
2. DSP Processor Fundamentals: Architectures & Features – Lapsley et al., S. Chand & Co. Digital Signal Processing Applications Using the ADSP-2100 Family, Prentice-Hall, Inc.

3. The Scientist and Engineer's Guide to Digital Signal Processing by Steven W. Smith, Digital Signal Processing, California Technical Publishing, San, Diego, CA.
4. Embedded Media Processing, David J. Katz and Rick Gentile, Elsevier, 2005.

KKR & KSR INSTITUTE OF TECHNOLOGY AND SCIENCES

(Autonomous)
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
Semester – IV-I

Course Code	Course Name	L T P C
23SH7T02	MANAGEMENT SCIENCE	2 0 0 2

Course Objectives:

1. The basic concepts, principles, and functions of management along with organizational structures and motivational theories.
2. The concepts and techniques related to operations, inventory, quality, and project management used in business and industries.
3. The fundamentals of Human Resource Management and Marketing Management including job evaluation, wage plans, and marketing strategies.

Course Outcomes:

CO1 To understand the basic concepts and functions of management and organization. — Understanding (L2)
CO2 To understand operations management techniques such as plant layout and work study. — Understanding (L2)
CO3 To apply inventory and quality control techniques in management problems. — Applying (L3)
CO4 To understand PERT and CPM techniques for project planning and scheduling. — Understanding (L2)
CO5 To understand the basics of Human Resource and Marketing Management. — Understanding (L2)

UNIT-I :

Concepts of Management and Organization: Nature and Scope of Management, Functions of Management, Levels of Management, Evolution of Management Thought – Taylor's Scientific Management, Fayol's 14 Principles of Management, Systems approach to Management, - Theories of Motivation – Maslow's need hierarchy theory, Theory X and Theory Y, Vrooms' expectations theory, Hawthrone Experiments.
Designing Organisation Structure: Basic Concepts related Organisation; Departmentation, Decentralisation, Types of organisation structures.

UNIT-II :

Operations Management: Plant Location– factors affecting the plant location, types of plant layouts.
Work study – Definition and objectives of Work Study, Process mapping tools of work study.
Difference between Micro-Motion Study and Memo – Motion Study. Work measurement – time study and Work Measurement, Therbligs.

Unit-III:

Material Management: Need & Objectives, Inventory Classification techniques – ABC & VED analysis). Techniques of Inventory control- EOQ, ABC analysis,(simple problems)
Statistical Quality Control - Control charts (P-chart, R-chart, and C-chart) Simple problems Acceptance Sampling and Total Quality Management – quality circles, ISO 9000 series.

UNIT-IV :

Project Management: (PERT/CPM): Development of Network – Difference between PERT and CPM Identifying Critical Path-Probability- Project Crashing (Simple Problems).

UNIT-V :

Introduction to Human Resource Management: Functions of HRM, Job Analysis, Techniques of Job Evaluation and Merit Rating, Wage payment plans (Simple Problems), Marketing Management- Functions of Marketing – Marketing strategies based on product Life Cycle.

TEXT BOOKS:

1. O.P. Khanna (2004), Industrial Engineering & Management, Dhanpat Rai, New Delhi.
2. VSP Rao, Human Resource Management, Vikas Publishing House, Uttar Pradesh.

REFERENCE BOOKS:

1. Stoner, Freeman (2005), Gilberth, Management, 6th edition, Pearson Education, New Delhi.
2. Panner Selvam (2004), Production and Operations Management, Prentice Hall of India, New Delhi.
3. Ralph M. Barnes (2004), Motion and Time studies, John wiley and Sons.
4. L.M. Prasad, Human Resource Management, Sultan Chand and Sons, New Delhi.